Measuring inflation in real time: Evidence from Bolivia*

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Executive Summary

For years, inflation has been measured using in-person surveying methods. Although robust, this data is often only available on a monthly basis. Given how quickly inflation can change, policymakers in emerging economies are often flying blind for weeks between inflation reports. Real-time consumer price indexes (CPIs) are now possible and relatively inexpensive with the rise of online retailing and price data. Indeed, Professors Alberto Cavallo and Roberto Rigobon developed web-scrape-derived CPIs for over 50 countries as a part of the Billion Prices Project [1]. Their work demonstrates that online prices track well with official data and can sometimes expose misreporting by statistical agencies. I apply a similar methodology to build a real-time retail index for the Bolivian economy. With a correlation of 0.67 between my index's monthly inflation and the matching official series, I demonstrate that emerging economies have a unique opportunity to not only improve their data collection methods but also potentially leapfrog developed nations such as the U.S., which continue to rely on traditional in-person methods and slower scanner data. Developing nation statistical agencies should invest in creating robust real-time CPIs using web-scraped prices as a reference for policy and the public.

Keywords: real-time inflation, web-scraped prices, consumer price index, Bolivia, emerging markets

1 Introduction

In mid-2025, Bolivia suffered its worst inflation episode in over 40 years. Prices rose by 20% year-over-year in 2025, and real GDP fell by 1.2% on an annual basis [2, 3]. The fall in output was primarily due to large real declines in commerce and mineral extraction [4]. The stagflationary environment contributed to a projected rise in the poverty rate (USD4.20 a day in 2021 PPP terms, the World Bank's lower-middle-income poverty line) from 4.9% in 2024 to 6.4% in 2025 [5]. Bolivians' suffering translated into a resounding defeat of the incumbent party, Movimiento al Socialismo (MAS), in the 2025 presidential election.

*I leveraged artificial intelligence to synthesize existing literature, code the data ingestion and calculation pipeline along with the webpage, and format and edit writing. Access the index here: `https://thomasriveros.github.io/robust-bolivia-cpi-website/`

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Given that inflation was, and remains, one of the primary concerns of the electorate, it is critical that policymakers understand the real-time effects of their legislation in order to properly orient fiscal and monetary policy. For example, when the newly elected Paz administration increased fuel prices to put an end to the shortages on December 17, 2025, many raised concerns about rising inflation. In theory, a one-time shock to fuel prices should immediately increase prices through a cost push channel. That is, production, transportation, and input costs rise due to the increased gas prices, which means less output for the same number of inputs. However, when looking at official data, we see that prices only rose by 0.59% month-over-month (MoM), 1.31%, and −0.62% in December, January, and February, respectively [2]. One likely reason for the muted impact is that the reduction in the subsidy also eliminated the fuel shortage the country experienced. After all, having no gas is more costly than having expensive gas. With higher frequency data, the Paz administration would not have needed to wait weeks until the publication of the December number to see that the subsidy removal did not greatly aggravate inflation (Figure 1).

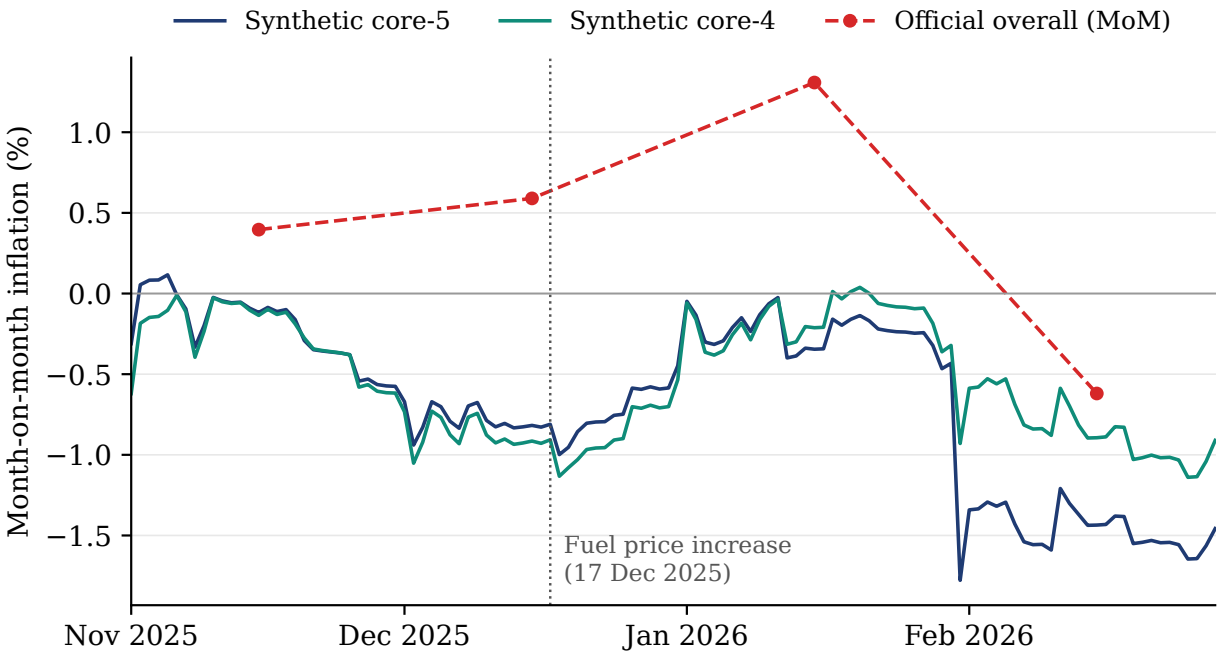


Figure 1: Synthetic inflation around the December 2025 fuel price increase
Source: Author's own work based on [6], accessed September 2026.

This policy scenario is not unique to Bolivia. Many developing nations face highly volatile social landscapes and could greatly benefit from higher frequency data. Indeed, although many countries, such as Australia [7] and the U.K. [8], leverage web-scraped and scanner data, none of their national statistical agencies publishes a daily CPI. Thus, emerging markets have a unique opportunity to leapfrog the developed nations' inflation estimates by building cheap, replicable, web-based CPIs. In this chapter I demonstrate the feasibility of creating such a CPI. For my data I draw on a GitHub repository created by Mauricio Foronda to access daily product-price observations from a major Bolivian supermarket chain [6]. I then apply a methodology similar to that of Cavallo (2013) to create a synthetic index [9].

2 Why real-time inflation matters in emerging economies

Inflation is a persistent issue in emerging economies. Often because of weak fiscal discipline, limited central bank independence, or an economy unable to absorb external shocks, prices rise more rapidly in emerging markets. From 2000 to 2025, the average annual inflation rate for emerging markets was 6.3%, while in advanced economies it stood at 2.2% [10]. Inflation also disproportionately impacts low-income households [11] since they derive most of their income from wages, pensions, or social benefits, which tend to be sticky (i.e., do not quickly adjust with inflation), and hold most of their savings in cash. Since the wealthier hold more real assets (such as company stock), they tend to be better shielded from the effects of inflation since companies can adjust prices. The overall result is a large decline in real incomes for the poorest.

Central to controlling inflation is the need to anchor inflation expectations. If households and firms believe prices will soon rise, they will preempt the price increase and purchase goods and services immediately, or in the case of the firms, raise their prices before others. This attempt to "beat" the inflation increases aggregate demand, which subsequently increases the price level. Individuals and firms generate their expectations, in part, from past inflation data. If people do not trust the government's inflation estimates, they will assume it is higher than the official numbers (since politicians have little incentive to over-report inflation). Thus, even if the inflation estimation is sound, distrust can actually raise inflation on its own. For example, Cavallo (2013) found inflation measured with online prices in Argentina to be nearly three times higher than the official data [9]. Since many other measures rely on inflation adjustments, poverty was chronically under-reported and real growth over-reported. Critically, Argentinians understood that inflation was much higher than official data and thus had to generate expectations with their own experience. The result was that inflation expectations were around 30% while reported inflation was 9.7% in 2007 [9]. Because inflation data was not easily auditable by the public, they lost confidence in the quality of the estimates. A daily updating, open-source price index would reduce the transparency problem. Citizens would be able to see and audit the data sources, calculations, and final index. Indeed, they could even calculate a personalized CPI based on their expenditure basket. In emerging markets, signalling credibility and accountability goes a long way in building confidence in the authorities.

In addition to the credibility signal, real-time data also allows policymakers to quickly move to combat rising prices. Inflation can accelerate within a few weeks, but the lag between a price shock and the policy response is often measured in months. A real-time measure means that fiscal and monetary policy can act in a countercyclical manner before expectations become completely unanchored and adjustment becomes far more costly.

3 The index

To create the index, I use data scraped by Mauricio Foronda from a major Bolivian retail chain with locations in Santa Cruz, La Paz, and Cochabamba (the three major Bolivian cities) [6]. The dataset consists of date, product name, price, category, subcategory, ID, and city variables. The government has 12 categories within which all goods and services fall, and the supermarket only has coverage for five of them (see Table 1). I begin by taking the price relative of each product, which

is the price of the product on the date of interest divided by the last available price of that product. For example, if the price of a kilogram of rice was 12.0 bolivianos (Bs) on January 1, 2025, and 12.6 Bs on January 2, 2025, then the price relative would be $12.6/12.0 = 1.05$. The price relative serves two purposes: firstly, to keep expensive items from overshadowing less expensive ones and secondly, to ensure the product is observed for at least two days before entering the index. If a product goes unobserved for more than 40 days, I drop its relative on the day it returns and treat its new price as the base. Next, I take the geometric mean of all the price relatives in a given category. In this example, the rice's price relative would be averaged with a bag of choclo's (corn) price relative and a cut of meat's price relative, since they are all in the food and non-alcoholic beverages category. I then chain each category's daily geometric means into an index starting at 100. Next, I take the weighted average of each category's index, using official INE weights rescaled so the five covered categories sum to one (see Table 1). This weighted average gives me the price index at the city level. I then apply city weights (Santa Cruz 41.47%, La Paz 39.10%, Cochabamba 19.43%) to get the national index. Finally, I measure synthetic inflation as the 30-day percentage change in the national index. Box 1 summarizes the full calculation.

Table 1: Covered expenditure categories and weights

Category	Official weight	Core-5 weight (%)	Core-4 weight (%)
Food and non-alcoholic beverages	27.06	55.08	65.10
Clothing and footwear	7.56	15.39	–
Miscellaneous goods and services	7.55	15.37	18.16
Furniture, household goods and services	6.08	12.38	14.63
Alcoholic beverages and tobacco	0.88	1.79	2.12
Share of official CPI covered		49.13	41.57

Notes: Core-5 and core-4 weights rescale the official INE weights of the covered categories to sum to 100. The remaining seven official categories are not sold by the supermarket chain.
Source: Author's own work based on [2], accessed September 2026.

To categorize the products, I use two methods for two separate indexes. The first, the "core-5" index, includes food and non-alcoholic beverages; clothing and footwear; miscellaneous goods and services; furniture, household goods and services; and alcoholic beverages and tobacco. For this index, I use a large language model (LLM) pipeline that sends products to Google Gemini 2.5 Flash to be categorized automatically. This method allows new products to be categorized as soon as they arrive. When I reviewed the output of the LLM-based classification, however, I found two problems. First, it miscategorized shoe polish as clothing and footwear when, in fact, it belongs in furniture, household goods and services. Because shoe polish made up the majority of products in clothing and footwear, too few genuine products remained in that category once it was reassigned, and the category was no longer stable enough to be included. Second, the LLM at times produced classifications outside the official INE categories. I therefore adopted a new pipeline in which I first map each supermarket subcategory to an INE category and fall back to AI classification only when needed. I report both the legacy "core-5" index and the updated "core-4" index, which excludes clothing and footwear and uses the new classification method. Because the two indexes differ in both basket and classification method, differences in their performance cannot be attributed to either change alone.

Box 1. Construction of the synthetic price index

Notation

i	product
c	city: Santa Cruz, La Paz, Cochabamba
k	expenditure category (core-5 or core-4 basket)
t	day
$p_{i,c,t}$	price of product i in city c on day t (bolivianos)
$\tau_{i,c}(t)$	last day before t on which product i was observed in city c
$S_{c,k,t}$	set of products in city c, category k with a valid relative on day t
$n_{c,k,t}$	number of products in $S_{c,k,t}$
ω_k	official INE weight of category k
s_c	weight of city c

Step 1: Price relatives. Each product's price is compared with its last observed price:

$$r_{i,c,t} = \frac{p_{i,c,t}}{p_{i,c,\tau_{i,c}(t)}}, \qquad \text{valid only if } t - \tau_{i,c}(t) \le 40. \tag{1}$$

If a product returns after more than 40 days, its relative on the day of return is dropped and its new price becomes the base.

Step 2: Elementary Jevons index. Within each city and category, take the geometric mean of the relatives:

$$J_{c,k,t} = \exp\left(\frac{1}{n_{c,k,t}} \sum_{i \in S_{c,k,t}} \ln r_{i,c,t} \right). \tag{2}$$

Step 3: Chaining. Link the daily geometric means into a price level that starts at 100:

$$I_{c,k,t} = I_{c,k,t-1} \times J_{c,k,t}, \qquad I_{c,k,0} = 100. \tag{3}$$

Step 4: Category weights. Rescale the official weights of the covered categories so they sum to one:

$$w_k = \frac{\omega_k}{\sum_{j \in \text{basket}} \omega_j}. \tag{4}$$

Step 5: City index.

$$I_{c,t} = \sum_{k \in \text{basket}} w_k \, I_{c,k,t}. \tag{5}$$

Step 6: National index.

$$I_t = \sum_c s_c \, I_{c,t}, \qquad s_{\text{SCZ}} = 0.4147,\ s_{\text{LPZ}} = 0.3910,\ s_{\text{CBBA}} = 0.1943. \tag{6}$$

Step 7: Synthetic inflation. The 30-day percentage change, compared with official month-over-month inflation:

$$\pi_t = 100 \times \left(\frac{I_t}{I_{t-30}} - 1 \right). \tag{7}$$

4 Results

Both the "core-5" and "core-4" indexes show strong correlation with actual inflation numbers (Table 2). The "core-5" index has a correlation coefficient of 0.68 when compared to the official equivalent index. Likewise, "core-4" stands at 0.67. City-level correlations vary widely for both indexes.

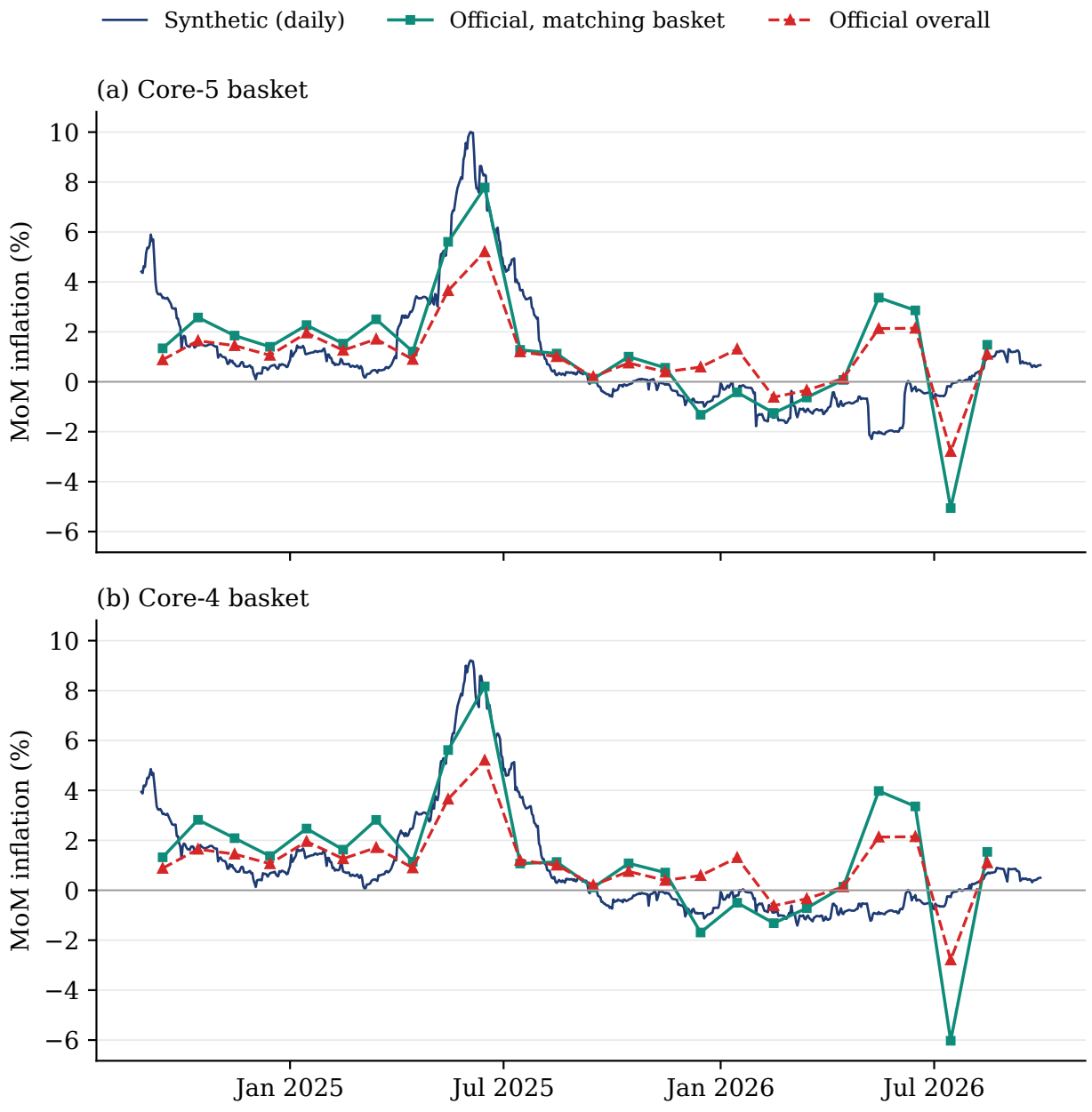


Figure 2: Synthetic and official month-over-month inflation, national
Source: Author's own work based on [6] and [2], accessed September 2026.

Table 2: Correlation and mean absolute deviation between synthetic and official monthly inflation

		Core-5		Core-4	
Comparison	N	Correlation	MAD	Correlation	MAD
National, matching official basket	24	0.676	1.356	0.668	1.437
National, overall official CPI	24	0.669	1.386	0.707	1.279
Santa Cruz	24	0.792	1.177	0.782	1.111
Cochabamba	24	0.624	1.559	0.640	1.621
La Paz	24	0.369	2.083	0.419	2.218

Notes: MAD = mean absolute deviation between synthetic and official MoM inflation, in percentage points.
Source: Author's own work based on [6] and [2], accessed September 2026.

La Paz only sees a correlation of 0.37 and 0.42, while Santa Cruz is 0.79 and 0.78 for "core-5" and "core-4," respectively. Mean absolute deviation for both indexes ranges from 1.1 to 2.2 percentage points.

The two indexes also appear to provide an early read on official inflation, as demonstrated by Table 3 with 0, 15, and 20 day lags. The slope between realized inflation and synthetic inflation falls from the zero-day to the 15-day lag and is roughly stable afterward. At a zero-day lag, a one-percentage-point increase in core-5 and core-4 inflation is associated with a ∼0.7 and ∼0.8 percentage-point increase in official equivalent inflation, respectively. At the 20-day lag (i.e., official equivalent inflation regressed on synthetic inflation 20 days before the middle of the month), the relationship is in the 0.6–0.8 range for both indexes. For official overall inflation (i.e., the full consumer price index), the relationship is weaker, ranging from 0.37 to 0.47 percentage points. All coefficients are significantly different from zero at the 0.05 level, and all but one at the 0.01 level. However, the regression output should be interpreted very carefully given the extremely small number of observations and the tendency of the synthetic index to not capture all price swings (such as the May–June 2026 blockade period).

Table 3: Regression of official CPI on synthetic price indices, by lag and estimator

		Regressor: synthetic Core-4					Regressor: synthetic Core-5				
Lag	Method	N	Slope	SE	p	Int.	N	Slope	SE	p	Int.
Panel A: Dependent variable = official CPI for the matching basket											
	OLS	24	0.795	0.189	0.0004	0.553	24	0.699	0.163	0.0003	0.638
0	OLS (HC1 SEs)	24	0.795	0.154	<0.0001	0.553	24	0.699	0.161	<0.0001	0.638
	Huber RLM	24	0.826	0.139	<0.0001	0.602	24	0.745	0.123	<0.0001	0.649
	OLS	24	0.683	0.196	0.0021	0.653	24	0.609	0.159	0.0009	0.730
15	OLS (HC1 SEs)	24	0.683	0.210	0.0012	0.653	24	0.609	0.167	0.0003	0.730
	Huber RLM	24	0.729	0.164	<0.0001	0.703	24	0.647	0.133	<0.0001	0.741
	OLS	23	0.679	0.215	0.0047	0.659	23	0.615	0.181	0.0028	0.720
20	OLS (HC1 SEs)	23	0.679	0.267	0.0108	0.659	23	0.615	0.224	0.0060	0.720
	Huber RLM	23	0.777	0.197	<0.0001	0.707	23	0.700	0.158	<0.0001	0.718
Panel B: Dependent variable = official overall CPI											
	OLS	24	0.465	0.099	0.0001	0.658	24	0.420	0.100	0.0003	0.723
0	OLS (HC1 SEs)	24	0.465	0.086	<0.0001	0.658	24	0.420	0.101	<0.0001	0.723
	Huber RLM	24	0.461	0.085	<0.0001	0.722	24	0.429	0.082	<0.0001	0.760
	OLS	24	0.398	0.105	0.0010	0.718	24	0.367	0.097	0.0010	0.777
15	OLS (HC1 SEs)	24	0.398	0.119	0.0008	0.718	24	0.367	0.108	0.0007	0.777
	Huber RLM	24	0.381	0.111	0.0006	0.791	24	0.373	0.091	<0.0001	0.821
	OLS	23	0.401	0.114	0.0021	0.726	23	0.374	0.110	0.0027	0.779
20	OLS (HC1 SEs)	23	0.401	0.147	0.0064	0.726	23	0.374	0.138	0.0069	0.779
	Huber RLM	23	0.412	0.114	0.0003	0.806	23	0.405	0.103	0.0001	0.822

Notes: Each cell reports a regression of official month-over-month inflation (panel) on the synthetic index's 30-day inflation (column block, Equation 7), with the synthetic series lagged by the number of days shown. Official monthly values are dated to the 15th of the reference month. In Panel A, the dependent variable is the official basket matching each column: official Core-4 for the Core-4 block and official Core-5 for the Core-5 block. OLS = ordinary least squares with conventional standard errors. HC1 = heteroskedasticity-consistent standard errors with small-sample correction. Huber RLM = robust linear model using Huber's M-estimator. SE = standard error; Int. = intercept.
Source: Author's own work based on [6] and [2], accessed September 2026.

5 Limitations and proposals

As demonstrated by the correlation and lagged regression results, the indexes are far from perfect. The "core-4" and "core-5" measures cover only 41.6% and 49.1% of the official weights, respectively, so if products outside the included categories move differently than those within them, the synthetic indexes will not capture true inflation. Perhaps more important is the May–June 2026 period, when, as a result of mass protests and blockades, MoM overall inflation hit 2.13% and 2.15%, respectively [2], while both synthetic indexes reported negative inflation (Figure 3). Coverage does not explain this gap: official inflation for the equivalent (core-4) basket rose even more, by 3.51% in May and 2.98% in June. The problem therefore lies in where prices are measured, not which categories are included. The supermarket chain may have absorbed the costs of the blockades or drawn down large stockpiles, while the informal markets passed supply shortages through to prices quickly. Products most affected by the blockades may also have gone out of stock and dropped out of the index, leaving only goods whose prices were not under pressure. Irrespective of the reason, the supermarket chain appears far less sensitive to blockade shocks than the markets most Bolivians shop in. Any real-time CPI should therefore draw on multiple types of outlets and acknowledge that it cannot fully track informal sector prices.

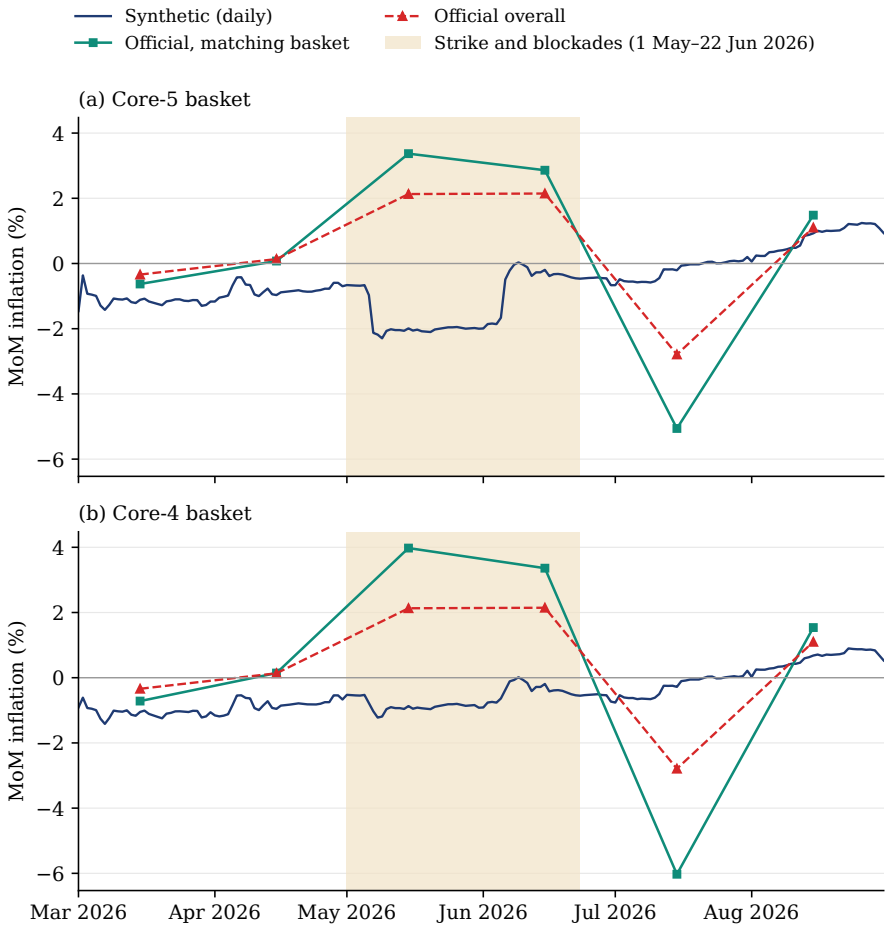


Figure 3: Synthetic and official inflation during the May–June 2026 blockades
Source: Author's own work based on [6] and [2], accessed September 2026.

Despite the imperfections, this chapter demonstrates that a real-time consumer price index built from online prices is feasible in an emerging market, even with limited data sources. Over 24 months, two synthetic indexes for Bolivia tracked official monthly inflation with correlations of 0.67 to 0.71 and gave an early signal weeks before the official release.

The case for such indexes is strongest where inflation is volatile and trust in official statistics is fragile. Publishing these CPIs would provide consumers and businesses with better information, improve institutional trust through auditability, and provide policymakers with real-time feedback on legislation. Because the tools are open-source and inexpensive, emerging-market statistical agencies can adopt them without waiting for advanced economies to lead.

Three recommendations follow.

1. **Build and publish.** National statistical offices, including INE, should build a daily online price index and publish it alongside the monthly CPI, with open code and data.
2. **Expand coverage first.** Before the index informs policy decisions, agencies should add retailers and outlet types, including informal markets, as well as categories and cities, prioritizing the cities where accuracy is weakest.
3. **Be transparent about accuracy.** Agencies should publish accuracy metrics with every release and present the index as an early-warning complement to the official CPI, not a substitute for it.

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